

CyberWatch – Enterprise Threat Intelligence Command Center

Cybersecurity Threat Monitoring & Risk Analytics Platform

POWER BI

POWER QUERY

DAX

EXCEL



The Cybersecurity Visibility Gap

Organizations generate massive network traffic daily , yet most lack the tools to detect, analyze & respond to threats.

1

Slow Threat Detection

Critical attacks go unidentified until damage is done

2

Fragmented Monitoring

No centralized view of suspicious activity across the enterprise

3

High-Risk Blind Spots

Severe threats buried in noise with no prioritization

4

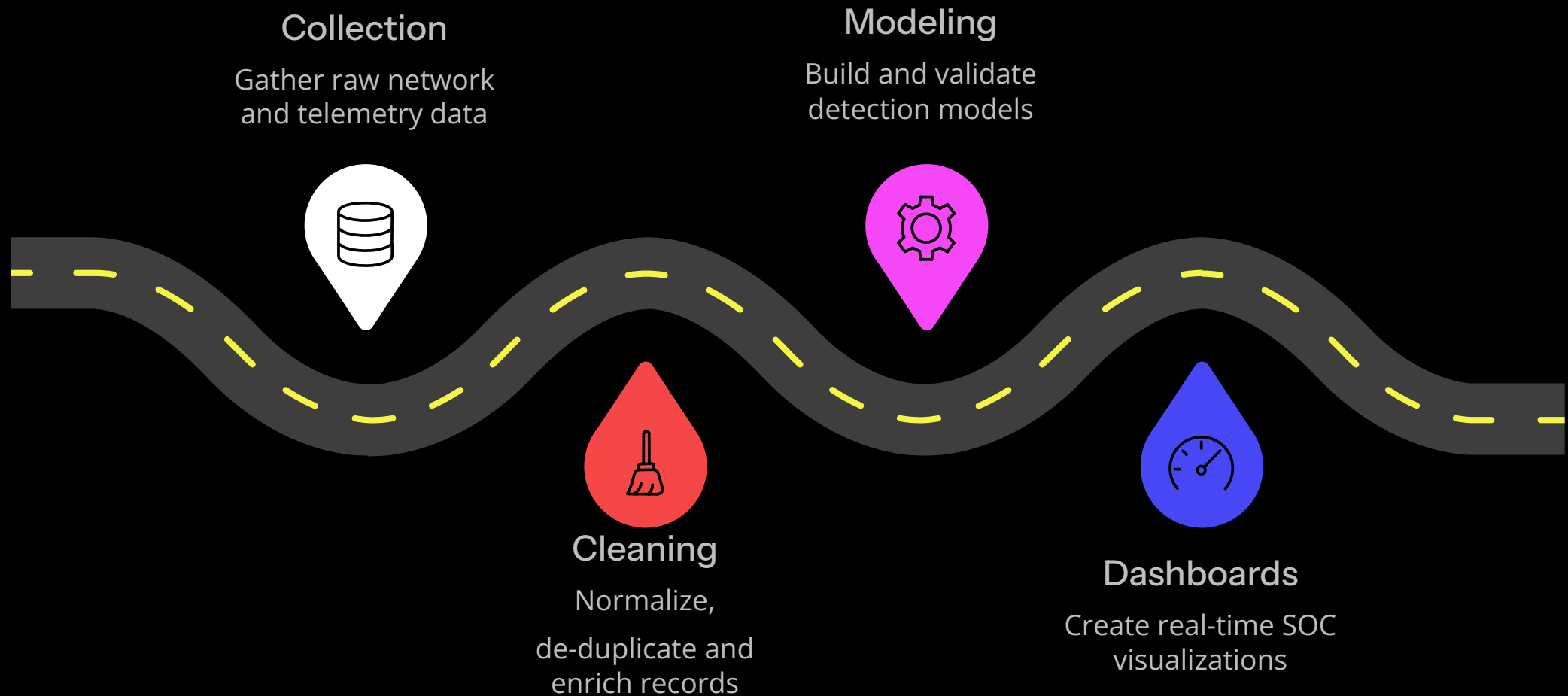
Inefficient Response

Incident teams lack actionable intelligence to act fast



From Raw Data to Real-Time Intelligence

CyberWatch follows a structured analytics pipeline — transforming raw network traffic into actionable SOC intelligence.



Each stage is engineered for enterprise-grade reliability, ensuring SOC teams receive clean, modeled & visualized threat intelligence at a glance.

Dataset & Data Preparation

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
id	dur	proto	service	state	spkts	dpkts	sbytes	dbytes	rate	sttl	dttl	sload	dload	sloss	dloss	sinpkt	dinpkt	sjit	djit
1	0.121478	tcp	-	FIN	6	4	258	172	74.08749	252	254	14158.94	8495.365	0	0	24.2956	8.375	30.17755	2
2	0.649902	tcp	-	FIN	14	38	734	42014	78.47337	62	252	8395.112	503571.3	2	17	49.915	15.43287	61.42693	13
3	1.623129	tcp	-	FIN	8	16	364	13186	14.17016	62	252	1572.272	60929.23	1	6	231.8756	102.7372	17179.59	1
4	1.681642	tcp	ftp	FIN	12	12	628	770	13.67711	62	252	2740.179	3358.622	1	3	152.8765	90.23573	259.0802	49
5	0.449454	tcp	-	FIN	10	6	534	268	33.37383	254	252	8561.499	3987.06	2	1	47.75033	75.6596	2415.838	2
6	0.380537	tcp	-	FIN	10	6	534	268	39.41798	254	252	10112.03	4709.135	2	1	39.92878	52.241	2223.73	6
7	0.637109	tcp	-	FIN	10	8	534	354	26.68303	254	252	6039.783	3892.584	2	1	68.26778	81.13771	4286.829	13
8	0.521584	tcp	-	FIN	10	8	534	354	32.59303	254	252	7377.527	4754.747	2	1	55.794	66.05414	3770.581	13
9	0.542905	tcp	-	FIN	10	8	534	354	31.31303	254	252	7087.796	4568.019	2	1	60.21089	68.109	4060.626	10
10	0.258687	tcp	-	FIN	10	6	534	268	57.98514	254	252	14875.12	6927.291	2	1	27.50511	39.1068	1413.686	5
11	0.304853	tcp	-	FIN	12	6	4142	268	55.76458	254	252	99641.47	5878.243	3	1	25.94882	53.6688	1471.649	80
12	2.093085	tcp	smtp	FIN	62	28	56329	2212	42.52097	62	252	211825.1	8152.559	28	8	34.31287	75.09245	3253.279	10
13	0.416952	tcp	-	FIN	10	6	534	268	35.97536	254	252	9228.88	4297.856	2	1	45.08878	64.4812	2610.908	99
14	0.996221	tcp	-	FIN	10	8	564	354	17.06449	254	252	4079.416	2489.407	2	1	110.6912	131.48	6542.815	2
15	0.576755	tcp	-	FIN	10	8	534	354	29.47525	254	252	6671.811	4299.919	2	1	64.08389	72.27157	4194.545	13
16	0.000002	udp	snmp	INT	2	0	138	0	500000	254	0	2.76E+08	0	0	0	0.002	0	0	0
17	0.728252	tcp	-	FIN	10	6	534	268	20.59727	254	252	5283.885	2460.687	2	1	80.91689	122.4622	5333.802	23
18	0.393556	tcp	http	FIN	10	8	860	1096	43.19589	62	252	15733.47	19494.05	2	2	43.72844	47.66915	2124.838	72
19	0.387852	tcp	-	FIN	10	6	534	268	38.67455	254	252	9921.31	4620.319	2	1	38.11967	68.2792	2226.183	10
20	0.53784	tcp	-	FIN	10	8	534	354	31.60791	254	252	7154.544	4611.037	2	1	59.66944	68.35186	3893.396	10
21	0.23372	tcp	-	FIN	10	6	534	268	64.17936	254	252	16464.14	7667.294	2	1	25.96889	40.812	1300.719	65
22	0.338017	tcp	http	FIN	10	6	998	268	44.37647	254	252	21277.04	5301.509	2	1	35.21811	56.5798	1928.551	82
23	0.964656	tcp	ftp	CON	14	12	690	950	25.91597	62	252	5315.885	7223.3	5	6	74.20431	59.70073	3869.615	65
24	0.449703	tcp	-	FIN	10	6	738	268	33.35535	254	252	11830.03	3984.852	2	1	42.61122	67.666	2352.174	90

What I Worked With

- Network traffic dataset with attack categories and risk levels
- Protocol, service and geographic metadata
- Event timestamps and activity classifications

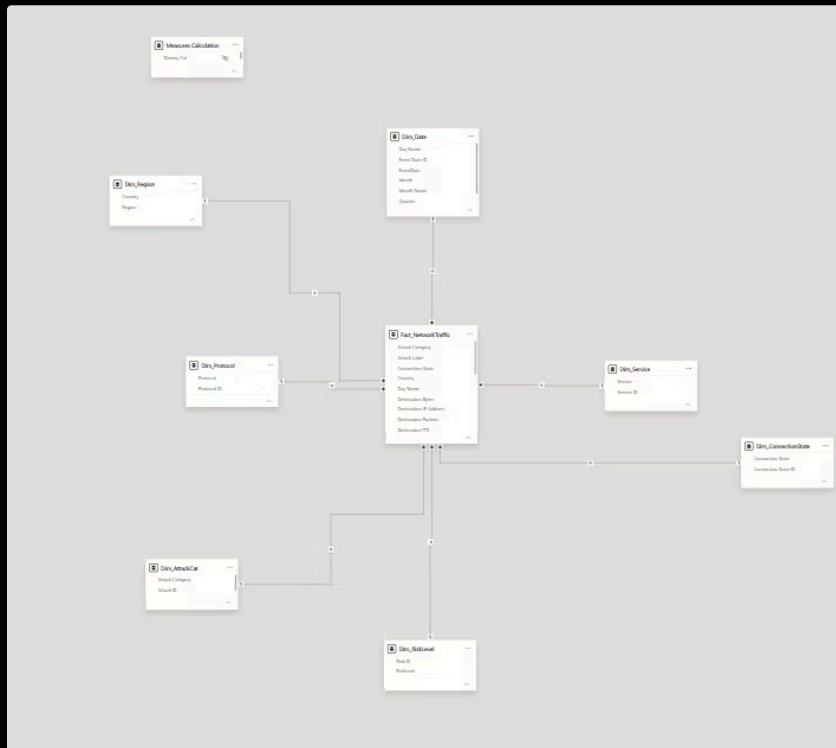
Power Query Transformations

- Missing value handling and null standardization
- Region mapping and geographic normalization
- Event type transformation and categorization

Star Schema Architecture & KPI Engineering

Analytics Architecture

A star schema connects a central **Fact_Attacks** table to dimension tables for Geography, Time, Service & Threat Type : enabling fast, accurate DAX calculations across the model.



Core KPI Measures

- **Total Attacks**
Aggregate threat count
- **Critical Threat Count**
Highest-severity incidents
- **Attack Rate %**
Proportion of malicious traffic
- **Risk Score**
Weighted severity index
- **Most Targeted Service**
Top exploited protocol

Enterprise Command Center — Four Intelligence Views

Four purpose-built dashboard pages deliver layered threat intelligence — from executive summaries to granular risk analysis.



Home Command Center

Executive threat summary with live KPIs & attack status



Threat Overview

Attack breakdowns by category, severity & timeline

Geo Intelligence

Geographic attack mapping with regional risk heatmaps



Risk Monitoring

Real-time risk scoring, trend analysis & alert prioritization

What the Data Revealed



DNS & HTTP Under Siege

Both services are the most heavily targeted attack vectors across the dataset



Late-Night Attack Window

High-risk attacks concentrated during off-hours — a classic evasion pattern



Asia & North America Hotspots

Both regions showed elevated attack activity, warranting focused SOC attention



Critical = Lower Frequency, Higher Severity

Critical threats were less frequent but carried the highest potential impact



Traffic Spikes Identified

Suspicious traffic surges detected during specific periods, indicating coordinated activity

Engineering Through Complexity

Challenges Faced

Missing Values

Incomplete network records required careful imputation

Data Standardization

Inconsistent formats across protocols and regions

Threat Simulation Realism

Balancing realistic attack patterns with clean data

KPI Engineering

Complex DAX logic for weighted risk scoring

Solutions Delivered

Power Query Cleaning
Null handling & Type Normalization

Standardized Schema
Unified data model with consistent field mappings

Curated Attack Profiles
Validated threat categories against known patterns

Modular DAX Measures
Reusable KPI logic with clear documentation

Enterprise Value Delivered

CyberWatch transforms how SOC teams see, prioritize & act on cyber threats : delivering measurable operational improvement.

4

Intelligence Views

Dashboards covering Home, Threat, Geo & Risk & Executive layers

6+

KPI Measures

Engineered DAX metrics for full threat visibility

1

Centralized Platform

Single source of truth for all SOC monitoring activity

5

Attack Categories

Classified and tracked across the full threat landscape



Stay Secure. Stay Ahead.

CyberWatch.

CyberWatch demonstrates how Power BI can be used to design a simulated enterprise cybersecurity monitoring platform for analyzing attack patterns, threat behavior, geographic risk trends & SOC-style security insights using synthetic network traffic data.

Future Enhancements

-  **Real-Time Streaming**
Live data integration for instant threat visibility
-  **SIEM Connectivity**
Native integration with enterprise security platforms
-  **AI Anomaly Detection**
Machine learning models for predictive threat analysis
-  **Automated Alerts**
Proactive notifications for critical incident response

